

# PolicyBench

Benchmarking no-tool tax-and-benefit estimation in frontier language models

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Source: [Article Notebook](#)

## 1 Abstract

PolicyBench evaluates whether frontier language models can estimate household tax and benefit outputs from household facts without tools. The benchmark covers the United States and the United Kingdom, uses sampled US Enhanced Current Population Survey (CPS)-derived scenarios and a public UK calibrated transfer dataset, and scores outputs with a bounded 0-100 metric. The manuscript snapshot is a 100-household-per-country public preview, not a protected held-out leaderboard: the current public scenario explorer exposes prompts and reference outputs, so open-set leakage is a central limitation of any public ranking. In the frozen snapshot used for this manuscript (2026-05-01), gpt-5.5 is top-scoring in the US, UK, and shared global views at 90.0, 77.2, and 83.6, respectively. In both countries, multi-step tax quantities and positive-dollar benefit cases are materially harder than zero cases. The live benchmark is available at <https://policybench.org>.

## 2 Introduction

Household tax-and-benefit estimation sits between arithmetic and policy discussion. Each case has numeric labels, but generating those labels requires filing status, household composition, income concepts, program thresholds, and jurisdiction-specific rules. PolicyBench measures whether models can map household records to those outputs.

Most current language-model evaluations do not test this mapping directly. General math benchmarks emphasize symbolic manipulation and exact final answers (Cobbe et al. 2021; Hendrycks et al. 2021), while generic question-answering benchmarks emphasize recall or instruction following. PolicyBench instead asks whether models can transform household facts into policy outputs without access to tools or simulators.

This matters for public-facing calculators, analyst workflows, and tax-preparation or screening systems. The benchmark is intended to separate verbal policy fluency from household-level quantitative prediction.

## 3 Related work

The most relevant prior work is tax and statutory reasoning. SARA frames statutory interpretation, including tax-relevant rule application, as a language-understanding problem (Holzenberger et al. 2021). LegalBench broadens this view and includes tax-oriented numeric

tasks such as `sara_numeric` (Guha et al. 2023). RuleArena evaluates rule-guided reasoning in regulation-style settings (Zhou et al. 2025), and TaxCalcBench evaluates frontier models on tax calculation from structured return-like inputs (Bock et al. 2025). Shanahan et al. report a similar split on the Volunteer Income Tax Assistance (VITA) test: models do better on tax knowledge questions than on open-ended calculation tasks (Shanahan et al. 2025).

PolicyBench also sits within a broader literature on quantitative reasoning benchmarks. Canonical math evaluations such as GSM8K and MATH normalized exact final-answer scoring for numeric tasks (Cobbe et al. 2021; Hendrycks et al. 2021). That norm is less natural for continuous-valued policy outputs, where being within 1 percent is closer than being off by an order of magnitude. Recent work has questioned pure exact-match scoring in numeric QA and temporal reasoning, arguing for error-aware metrics instead (Abbood et al. 2025). PolicyBench follows that direction by combining exactness with near-miss thresholds rather than collapsing everything into exact match alone.

Finance-domain benchmarks provide another relevant precedent. FinQA and TAT-QA both show that realistic quantitative reasoning often requires combining structured numbers with domain knowledge rather than solving stylized school-math problems (Chen et al. 2021; Zhu et al. 2021). PolicyBench differs in that its reference outputs are generated by executable tax-benefit microsimulation rather than annotated document questions, but the underlying motivation is similar: domain-specific quantitative reasoning deserves dedicated evaluation.

Finally, PolicyBench depends on two infrastructure literatures that are not themselves large language model (LLM) benchmarks. One is structured-output reliability, since the benchmark relies on multi-output JSON responses and parse coverage rather than free-form prose (Shorten et al. 2024). The other is tax-benefit microsimulation, where systems such as EUROMOD provide the methodological precedent for evaluating policy rules over household microdata (Sutherland and Figari 2013). There is also operational work on applying and evaluating AI systems in public-benefits settings, including caseworker-assist and Supplemental Nutrition Assistance Program (SNAP)-focused evaluations (Nava Labs 2026, 2025; ZenML LLMOps Database 2025). PolicyBench combines these strands into a public cross-model benchmark over household-level tax and benefit outputs.

## 4 Benchmark design

PolicyBench asks models to predict all benchmark outputs for a household in a single structured response. One response per household reduces repeated prompt cost and keeps the task at the household-return level rather than turning it into a sequence of unrelated one-output calls.

The benchmark uses a bounded 0-100 score. For amount variables, the score averages four hit rates:

1. exact

2. within 1%
3. within 5%
4. within 10%

For binary outputs, the score is exact accuracy. This keeps 100 as the ceiling while still giving partial credit for near misses on amount outputs. PolicyBench also tracks mean absolute error and related error metrics, but those are secondary to the bounded score. This choice preserves exact-match comparability while avoiding the failure mode that recent numeric-evaluation papers have criticized (Abbood et al. 2025).

All requested US outputs are annual amounts or annual eligibility indicators for tax year 2026; UK outputs are annual amounts or annual eligibility indicators for fiscal year 2026-27. For currency amounts, the “exact” hit rate means within 1 currency unit of the reference value after numeric parsing. Percentage-threshold hit rates use relative error when the reference value is nonzero and the same 1 currency-unit absolute tolerance when the reference value is zero. Binary outputs are parsed as 0 or 1 eligibility flags and scored by exact classification accuracy. Missing or unparseable numeric answers receive zero score for that requested output.

Aggregation proceeds in four steps. First, each household-output prediction receives a 0-100 score. Second, person-level coverage rows are averaged to their program group. Third, output groups are averaged with equal weight within each country. Fourth, global scores average the US and UK country scores for models run in both countries. The global score is therefore an equal-country summary for this benchmark sample, not a universally authoritative measure of tax-benefit competence. Household-equal impact scores and other alternative views are reported as sensitivity checks.

The benchmark requires each model response to include numeric answers and one explanation per requested output. The headline score uses only the numeric answers. Explanations are retained for scenario exploration and qualitative error analysis; they should not be interpreted as faithful traces of model reasoning.

Because explanations are required, the canonical task measures policy estimation under a public-facing structured-response contract, not isolated arithmetic accuracy. Prompt fairness is part of the benchmark contract. The current release uses one prompt template per country, with no model-specific tuning. Models receive the same household facts, the same requested outputs, and no web or tool access. The prompt states that unlisted numeric inputs are 0, unlisted boolean or status facts are false, and household characteristics are constant over the tax-benefit year. Provider-specific differences are limited to the response transport needed to obtain structured output.

#### 4.1 Frozen snapshot and open-set status

Leaderboard positions are version-sensitive, so manuscript claims refer to the frozen dashboard export in Table 1 rather than to the live site. The local run labels identify the source runs used

to create that export; the committed export is the manuscript artifact. The public site exposes the current prompts, predictions, explanations, and reference outputs for transparency. This makes the public leaderboard open-set: models, model providers, or benchmark users could learn from the released cases before later runs. Protected leaderboard claims would require a separate held-out or rotating set.

Source: [Article Notebook](#)

Table 1: Frozen manuscript snapshot.

|    | Item                                | Value                                                          |
|----|-------------------------------------|----------------------------------------------------------------|
| 0  | Snapshot date                       | 2026-05-01                                                     |
| 1  | Policy period                       | US tax year 2026; UK fiscal year 2026-27                       |
| 2  | Frozen export                       | app/src/data.json                                              |
| 3  | Frozen export SHA-256 prefix        | d3ee592e809d                                                   |
| 4  | Snapshot manifest                   | paper/snapshot/20260501/manifest.json                          |
| 5  | US scenarios SHA-256 prefix         | 46bc6b1ba56e                                                   |
| 6  | US reference outputs SHA-256 prefix | 21f09d04552d                                                   |
| 7  | UK scenarios SHA-256 prefix         | cc072e32df04                                                   |
| 8  | UK reference outputs SHA-256 prefix | d12f3f375feb                                                   |
| 9  | Benchmark spec SHA-256 prefix       | 532e15ba19bf                                                   |
| 10 | Prompt template SHA-256 prefix      | 3b81cd8ac0af                                                   |
| 11 | Scoring code SHA-256 prefix         | 39d47179c02f                                                   |
| 12 | US run label                        | us_100_stable_models_20260501_062249                           |
| 13 | UK run label                        | uk_100_hicbc_fixed_20260501_083402                             |
| 14 | Household sample seed               | 42                                                             |
| 15 | PolicyEngine-US                     | policyengine-us 1.687.0                                        |
| 16 | PolicyEngine-UK                     | policyengine-uk 2.88.13                                        |
| 17 | US data bundle                      | policyengine-us-data 1.73.0                                    |
| 18 | UK runtime bundle                   | policyengine-uk-data 1.40.4                                    |
| 19 | UK scenario source                  | enhanced_cps_2025.h5 public UK calibrated transfer artifact    |
| 20 | UK scenario-source SHA-256          | 199ebc61d29231b4799ad337a95393765b5fb5aede1834b93ff2acecced    |
| 21 | Households                          | 100 US and 100 UK                                              |
| 22 | Models                              | 12 shared models                                               |
| 23 | Output groups                       | 19 US and 7 UK                                                 |
| 24 | Condition                           | No tools, no web access, one structured response per household |
| 25 | Response contract                   | Numeric answer and non-empty explanation for every requested o |

Source: [Article Notebook](#)

Table 2: Model run configuration in the fr

|    | Model                         | Provider ID                          | US parsed   | UK parsed | Structured |
|----|-------------------------------|--------------------------------------|-------------|-----------|------------|
| 0  | gpt-5.5                       | gpt-5.5                              | 2,180/2,180 | 700/700   | provider s |
| 1  | gemini-3.1-pro-preview        | gemini/gemini-3.1-pro-preview        | 2,180/2,180 | 700/700   | provider s |
| 2  | grok-4.20                     | xai/grok-4.20-reasoning              | 2,180/2,180 | 700/700   | provider s |
| 3  | grok-4.3                      | xai/grok-4.3                         | 2,180/2,180 | 700/700   | provider s |
| 4  | gemini-3-flash-preview        | gemini/gemini-3-flash-preview        | 2,180/2,180 | 700/700   | provider s |
| 5  | claude-opus-4.7               | claude-opus-4-7                      | 2,180/2,180 | 700/700   | provider s |
| 6  | claude-sonnet-4.6             | claude-sonnet-4-6                    | 2,180/2,180 | 700/700   | provider s |
| 7  | gemini-3.1-flash-lite-preview | gemini/gemini-3.1-flash-lite-preview | 2,180/2,180 | 700/700   | provider s |
| 8  | grok-4.1-fast                 | xai/grok-4-1-fast-non-reasoning      | 2,180/2,180 | 700/700   | provider s |
| 9  | claude-haiku-4.5              | claude-haiku-4-5-20251001            | 2,180/2,180 | 700/700   | provider s |
| 10 | gpt-5.4-mini                  | gpt-5.4-mini                         | 2,180/2,180 | 700/700   | provider s |
| 11 | gpt-5.4-nano                  | gpt-5.4-nano                         | 2,180/2,180 | 700/700   | provider s |

Source: [Article Notebook](#)

## 5 Data and scenario construction

### 5.1 United States

The US benchmark is built from Enhanced Current Population Survey (CPS)-derived households using PolicyEngine US. The sampled households are filtered to keep a single-tax-unit structure while retaining variation in filing status, household composition, and income sources. Prompts include nonzero promptable raw inputs across relevant entities rather than a hand-curated summary, so the models see many of the same facts the simulator receives.

The current US release evaluates 19 output groups spanning federal income tax, refundable credits, payroll and self-employment tax, state and local income tax, Supplemental Nutrition Assistance Program (SNAP), Supplemental Security Income (SSI), Temporary Assistance for Needy Families (TANF), Affordable Care Act (ACA) premium tax credits, school-meal eligibility, and person-level coverage eligibility for the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), Medicaid, the Children’s Health Insurance Program (CHIP), Medicare, Head Start, and Early Head Start.

The output scope is intentionally narrower than the full PolicyEngine model. Table 3 summarizes the inclusion rule. The benchmark asks for WIC eligibility rather than a WIC dollar amount; WIC dollar values are used only as impact-weight proxies for coverage flags, not as requested model outputs.

Source: [Article Notebook](#)

|   | Scope decision          | Rationale                                                                           |
|---|-------------------------|-------------------------------------------------------------------------------------|
| 0 | Included                | Direct tax, credit, benefit, health-support, and coverage outputs that a household  |
| 1 | Excluded                | Intermediate tax bases, payroll subcomponents, and outputs that mainly require      |
| 2 | ACA Premium Tax Credit  | Retained as a deliberate health-support output; when local benchmark premium        |
| 3 | Binary coverage outputs | Requested as 0/1 eligibility flags and scored as classification tasks; their dollar |
| 4 | WIC                     | The benchmark asks for person-level WIC eligibility. It does not ask models to      |

Source: [Article Notebook](#)

## 5.2 United Kingdom

The UK benchmark is built from a calibrated public transfer dataset scored through PolicyEngine UK. The current public build starts from a public export of benchmark-compatible households from PolicyEngine US Enhanced CPS, maps those records into UK-facing inputs, and recalibrates them to selected UK targets. This creates a public UK-policy transfer benchmark without publishing restricted household microdata; it is not a representative evaluation over native UK household records and should not be treated as a substitute for Family Resources Survey (FRS)-based UK microdata. The current UK release evaluates seven outputs: Income Tax, National Insurance, Capital Gains Tax, Child Benefit, Universal Credit, Pension Credit, and Personal Independence Payment (PIP). Outputs that depend on status or award facts use prompt-visible facts rather than hidden take-up labels; for example, PIP-positive cases list the daily living and mobility award components used by PolicyEngine.

The UK data path is more synthetic than the enhanced FRS pipeline and inherits limitations from cross-country transfer, calibration choices, and the subset of variables that can be made prompt-visible. It supports the current public cross-country benchmark, but it is not equivalent to an enhanced-FRS-based benchmark and should not be used to make population-representative claims about UK households (Sutherland and Figari 2013).

## 5.3 Reference-output credibility

PolicyBench treats PolicyEngine outputs as benchmark reference outputs, not as administrative records. The reference source is nevertheless stronger than an ad hoc answer key: PolicyEngine is open source, used for household calculators and reform analysis, and externally checked in specific domains. In the UK, No. 10's data science team adapted PolicyEngine's open-source microsimulation model for experimental policy simulation, with validation against external projections before use (Woodruff 2026; Ghenis 2026). In the US, PolicyEngine's state tax modelling has been validated against the National Bureau of Economic Research (NBER) TAXSIM model, with reported penny-level agreement for the vast majority of 2021 test cases

(PolicyEngine 2024; Feenberg and Coutts 1993). PolicyEngine has also signed a memorandum of understanding with the Federal Reserve Bank of Atlanta for future validation work against the Atlanta Fed’s Policy Rules Database (Ghenis and Makarchuk 2025; Federal Reserve Bank of Atlanta 2026). The Atlanta Fed sources are a caveat rather than evidence of completed validation for this benchmark: they document planned collaboration and the comparison source, not finished checks of the frozen PolicyBench outputs. Taken together, these sources support using PolicyEngine as a transparent reference implementation with partial external validation, but they do not validate every benchmark output.

This does not make PolicyEngine infallible. During benchmark development, we performed a manual, developer-led discrepancy review, with LLM assistance used to triage and summarize candidate cases surfaced by model explanations. Table 4 summarizes the main discrepancy classes reviewed before the frozen snapshot. In those reviewed classes, discrepancies reflected model mistakes, prompt ambiguity later corrected, or upstream data/model issues fixed before the frozen snapshot; the reviewed discrepancy classes did not identify unresolved PolicyEngine reference-output defects. This is a qualitative development audit, not an independent sampled validation study or exhaustive validation of every reference value.

Source: [Article Notebook](#)

Table 4: Development discrepancy review

|   | Discrepancy class              | Review outcome                                                          |
|---|--------------------------------|-------------------------------------------------------------------------|
| 0 | US federal credits             | Reviewed cases where explanations omitted EITC or refundable CTC;       |
| 1 | US SNAP, SSI, and Medicaid     | Reviewed false-zero and false-positive explanations; common model fail  |
| 2 | US payroll and overtime        | Reviewed overtime-related cases during development; upstream data/m     |
| 3 | UK Child Benefit               | Reviewed HICBC-related misses; the benchmark now asks for gross Ch      |
| 4 | UK National Insurance          | Reviewed cases where models subtracted pension contributions or used    |
| 5 | UK UC, Pension Credit, and PIP | Reviewed positive-case misses and false positives; explanations usually |

Source: [Article Notebook](#)

## 6 Results

### 6.1 United States leaderboard

The US leaderboard for the frozen manuscript snapshot is shown in Table 5. The top three models in that snapshot are gpt-5.5 (90.0), grok-4.20 (89.3), and gemini-3.1-pro-preview (88.2).

Source: [Article Notebook](#)



Table 5: Top US benchmark models in the frozen manuscript snapshot.

|   | Model                  | Score | Exact | Within 10% | Parsed | Total |
|---|------------------------|-------|-------|------------|--------|-------|
| 0 | gpt-5.5                | 90.0  | 86.6  | 93.1       | 2180   | 2180  |
| 1 | grok-4.20              | 89.3  | 86.9  | 92.1       | 2180   | 2180  |
| 2 | gemini-3.1-pro-preview | 88.2  | 86.0  | 90.9       | 2180   | 2180  |
| 3 | grok-4.3               | 88.2  | 86.4  | 90.3       | 2180   | 2180  |
| 4 | gemini-3-flash-preview | 86.9  | 85.0  | 89.5       | 2180   | 2180  |

Source: [Article Notebook](#)

## 6.2 United Kingdom leaderboard

The UK leaderboard for the frozen manuscript snapshot is shown in Table 6. The top three models in that snapshot are gpt-5.5 (77.2), gemini-3.1-pro-preview (76.2), and grok-4.20 (75.1).

Source: [Article Notebook](#)

Table 6: Top UK benchmark models in the frozen manuscript snapshot.

|   | Model                  | Score | Exact | Within 10% | Parsed | Total |
|---|------------------------|-------|-------|------------|--------|-------|
| 0 | gpt-5.5                | 77.2  | 68.7  | 88.0       | 700    | 700   |
| 1 | gemini-3.1-pro-preview | 76.2  | 68.7  | 87.1       | 700    | 700   |
| 2 | grok-4.20              | 75.1  | 68.3  | 85.3       | 700    | 700   |
| 3 | grok-4.3               | 74.5  | 69.4  | 83.6       | 700    | 700   |
| 4 | gemini-3-flash-preview | 73.1  | 67.6  | 80.6       | 700    | 700   |

Source: [Article Notebook](#)

## 6.3 Global shared-model leaderboard

The global leaderboard averages US and UK scores for the models that have been run in both countries. In the frozen manuscript snapshot, the top three are gpt-5.5 (83.6), gemini-3.1-pro-preview (82.2), and grok-4.20 (82.2). The global table is an equal-country summary for this two-country benchmark sample; it should be read alongside the country tables and sensitivity views rather than as a universally authoritative ordering.

Source: [Article Notebook](#)

Table 7: Global shared-model leaderboard across US and UK runs.

|   | Model                         | Score | Exact | Within 10% | Parsed | Total |
|---|-------------------------------|-------|-------|------------|--------|-------|
| 0 | gpt-5.5                       | 83.6  | 77.7  | 90.6       | 2880   | 2880  |
| 1 | gemini-3.1-pro-preview        | 82.2  | 77.4  | 89.0       | 2880   | 2880  |
| 2 | grok-4.20                     | 82.2  | 77.6  | 88.7       | 2880   | 2880  |
| 3 | grok-4.3                      | 81.3  | 77.9  | 87.0       | 2880   | 2880  |
| 4 | gemini-3-flash-preview        | 80.0  | 76.3  | 85.1       | 2880   | 2880  |
| 5 | claude-opus-4.7               | 79.1  | 75.8  | 83.8       | 2880   | 2880  |
| 6 | claude-sonnet-4.6             | 78.9  | 76.2  | 81.9       | 2880   | 2880  |
| 7 | gemini-3.1-flash-lite-preview | 78.7  | 76.5  | 82.1       | 2880   | 2880  |
| 8 | grok-4.1-fast                 | 76.5  | 75.2  | 77.9       | 2880   | 2880  |
| 9 | claude-haiku-4.5              | 76.1  | 74.7  | 77.7       | 2880   | 2880  |

Source: [Article Notebook](#)

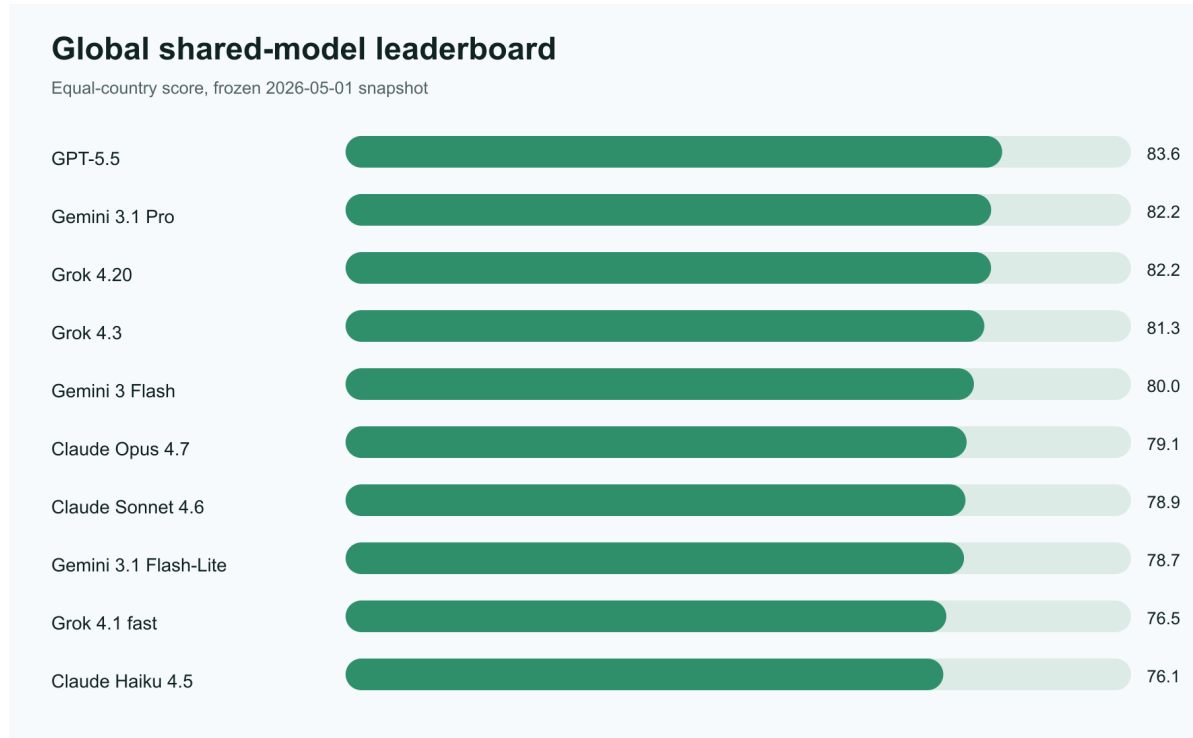


Figure 1: Global shared-model leaderboard in the frozen manuscript snapshot.

## 6.4 Simple baselines

Simple baselines help interpret score levels in a zero-heavy benchmark. Table 8 reports an always-zero response and a median-reference-by-output response on the same frozen household sample. These are not model competitors; they show how much of the bounded score can be earned without household-specific policy calculation.

Source: [Article Notebook](#)

Table 8: Simple baseline scores on the frozen manuscript snapshot.

|   | Baseline                   | Global score | US score | UK score |
|---|----------------------------|--------------|----------|----------|
| 0 | Always zero                | 77.7         | 81.9     | 73.6     |
| 1 | Median reference by output | 70.4         | 76.4     | 64.3     |

Source: [Article Notebook](#)

## 6.5 Sensitivity to benchmark view

The primary leaderboard gives each country equal weight globally and each output group equal weight within a country. Table 9 reports several alternative views. The top model is stable across the main global, amount-only, positive-reference-case, US-only, UK-only, and household-equal impact views. The zero-reference-case view changes the top rank, and lower ranks move across several sensitivity views. The global score is therefore best read as a summary statistic whose interpretation depends on these sensitivity checks.

Source: [Article Notebook](#)

Table 9: Leaderboard sensitivity to alternative scoring views.

|   | View                          | Rank 1                        | Rank 2                        | Rank 3                        |
|---|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| 0 | Main global leaderboard       | gpt-5.5 (83.6)                | gemini-3.1-pro-preview (82.2) | grok-4.20 (78.8)              |
| 1 | Amount outputs only           | gpt-5.5 (80.6)                | grok-4.20 (78.8)              | gemini-3.1-pro-preview (76.2) |
| 2 | Positive reference cases only | gpt-5.5 (43.8)                | gemini-3.1-pro-preview (36.9) | grok-4.20 (36.9)              |
| 3 | Zero reference cases only     | grok-4.3 (96.9)               | gemini-3.1-pro-preview (96.3) | gpt-5.5 (96.3)                |
| 4 | US only                       | gpt-5.5 (90.0)                | grok-4.20 (89.3)              | gemini-3.1-pro-preview (89.3) |
| 5 | UK only                       | gpt-5.5 (77.2)                | gemini-3.1-pro-preview (76.2) | grok-4.20 (76.2)              |
| 6 | US binary coverage only       | gemini-3.1-pro-preview (98.5) | grok-4.20 (98.5)              | grok-4.3 (98.5)               |
| 7 | Household-equal impact score  | gpt-5.5 (55.7)                | gemini-3.1-pro-preview (51.1) | grok-4.20 (51.1)              |

Source: [Article Notebook](#)

## 6.6 Sampling uncertainty

The manuscript snapshot uses 100 households per country, so small score gaps should not be overinterpreted. Table 10 reports a household-resampling bootstrap over the frozen sample using 1,000 percentile bootstrap draws. It resamples households within each country, recomputes country scores with equal output-group weighting, then recomputes the equal-country global score. The table is a sampling-uncertainty check for this household draw, not uncertainty over future model releases or prompt variants.

Source: [Article Notebook](#)

Table 10: Household-resampling uncertainty for the global leaderboard.

|   | Model                         | Score | 95% bootstrap score interval | Bootstrap rank range |
|---|-------------------------------|-------|------------------------------|----------------------|
| 0 | gpt-5.5                       | 83.6  | 82.3-84.9                    | 1-1                  |
| 1 | gemini-3.1-pro-preview        | 82.2  | 80.7-83.6                    | 2-4                  |
| 2 | grok-4.20                     | 82.2  | 80.9-83.5                    | 2-4                  |
| 3 | grok-4.3                      | 81.3  | 79.9-82.7                    | 2-4                  |
| 4 | gemini-3-flash-preview        | 80.0  | 78.6-81.4                    | 5-8                  |
| 5 | claude-opus-4.7               | 79.1  | 77.6-80.5                    | 5-8                  |
| 6 | claude-sonnet-4.6             | 78.9  | 77.5-80.3                    | 5-8                  |
| 7 | gemini-3.1-flash-lite-preview | 78.7  | 77.3-80.1                    | 5-8                  |
| 8 | grok-4.1-fast                 | 76.5  | 74.9-78.0                    | 9-12                 |
| 9 | claude-haiku-4.5              | 76.1  | 74.5-77.6                    | 9-12                 |

Source: [Article Notebook](#)

## 6.7 Hardest benchmark targets

The lowest-scoring US variables are multi-step tax quantities and sparse positive-dollar benefits. As shown in Table 11, federal income tax before refundable credits, state income tax before refundable credits, payroll tax, and SNAP are the hardest US outputs in the frozen manuscript snapshot.

Source: [Article Notebook](#)

Table 11: Hardest US variables by bounded score.

|    | Variable                                     | Score | Exact | Within 10% |
|----|----------------------------------------------|-------|-------|------------|
| 0  | federal_income_tax_before_refundable_credits | 42.6  | 37.5  | 50.2       |
| 16 | state_income_tax_before_refundable_credits   | 47.8  | 39.9  | 57.7       |

Table 11: Hardest US variables by bounded score.

|    | Variable                   | Score | Exact | Within 10% |
|----|----------------------------|-------|-------|------------|
| 4  | payroll_tax                | 62.8  | 50.7  | 75.7       |
| 14 | snap                       | 67.4  | 64.1  | 70.5       |
| 1  | federal_refundable_credits | 83.2  | 82.3  | 84.2       |

Source: [Article Notebook](#)

The same pattern appears in the UK run. Income tax and National Insurance are the two hardest UK outputs, while zero-heavy benefits and capital gains tax score higher overall.

Source: [Article Notebook](#)

Table 12: Hardest UK variables by bounded score.

|   | Variable           | Score | Exact | Within 10% |
|---|--------------------|-------|-------|------------|
| 2 | income_tax         | 36.7  | 25.8  | 54.8       |
| 3 | national_insurance | 48.9  | 39.5  | 62.3       |
| 5 | pip                | 74.0  | 74.0  | 74.0       |
| 6 | universal_credit   | 81.5  | 80.0  | 83.4       |
| 1 | child_benefit      | 84.2  | 76.8  | 92.0       |

Source: [Article Notebook](#)

## 6.8 Zero and positive cases

Overall hit rates can overstate performance on sparse programs because correct zeros are common. Table 13 and Table 14 therefore report within-10% performance for all cases, positive-reference cases, and zero-reference cases. The gap is large for several outputs: models often identify that a program or tax does not apply, but miss the amount when the reference value is positive.

Source: [Article Notebook](#)

Table 13: US within-10% performance by zero versus positive reference cases.

|   | Variable                                     | All cases | Positive cases | Zero cases |
|---|----------------------------------------------|-----------|----------------|------------|
| 0 | federal_income_tax_before_refundable_credits | 50.2      | 21.0           | 95.7       |
| 1 | state_income_tax_before_refundable_credits   | 57.7      | 30.5           | 96.7       |
| 2 | snap                                         | 70.5      | 18.1           | 100.0      |

Table 13: US within-10% performance by zero versus positive reference cases.

|   | Variable                   | All cases | Positive cases | Zero cases |
|---|----------------------------|-----------|----------------|------------|
| 3 | payroll_tax                | 75.7      | 67.0           | 96.8       |
| 4 | federal_refundable_credits | 84.2      | 24.4           | 93.2       |
| 5 | state_refundable_credits   | 85.4      | 5.1            | 97.4       |
| 6 | person_medicaid_eligible   | 85.7      | 62.1           | 94.9       |
| 7 | ssi                        | 91.9      | 26.7           | 99.2       |

Source: [Article Notebook](#)

Table 14: UK within-10% performance by zero versus positive reference cases.

|   | Variable           | All cases | Positive cases | Zero cases |
|---|--------------------|-----------|----------------|------------|
| 0 | income_tax         | 54.8      | 41.3           | 86.1       |
| 1 | national_insurance | 62.3      | 38.9           | 97.5       |
| 2 | pip                | 74.0      | NaN            | 74.0       |
| 3 | universal_credit   | 83.4      | 19.0           | 97.6       |
| 4 | child_benefit      | 92.0      | 68.9           | 98.5       |
| 5 | capital_gains_tax  | 92.6      | 19.4           | 99.8       |
| 6 | pension_credit     | 92.8      | 1.4            | 98.6       |

Source: [Article Notebook](#)

## 7 Failure modes

The benchmark surfaces a few recurring failure patterns.

First, models miss positive tax and benefit quantities more often than zero cases. In the US, federal income tax before refundable credits, state income tax before refundable credits, payroll tax, and SNAP are the four lowest-scoring outputs. These require the model to choose the right income concepts, exclusions, program thresholds, and sequencing before applying any final subtraction.

Second, the UK benchmark shows the same split between tax calculations and many benefit outputs. Income tax and National Insurance score below the benefit outputs in the frozen manuscript snapshot. Positive Universal Credit and Pension Credit cases remain difficult, so the result should not be read as a general claim that benefits are easy.

Third, structured-output reliability can affect rankings. The current manuscript snapshot has full parse coverage for the included models, but the benchmark still tracks coverage because



Figure 2: Output-level performance on zero-reference and positive-reference cases.

failures to return parseable numeric values should count as benchmark failures (Shorten et al. 2024).

## 8 Limitations

PolicyBench is not a substitute for a production tax-and-benefit calculator. Several caveats matter:

- model rankings can reflect output-contract reliability as well as policy reasoning
- zero-heavy outputs require separate positive-case interpretation
- the scoring rule is one benchmark view and should be checked against alternative metrics
- the global score is an equal-country summary for models run in both countries, not a universal ranking of model quality
- the public UK calibrated transfer dataset is not equivalent to enhanced Family Resources Survey quality and is not population-representative UK microdata
- the public scenario explorer exposes the current test set and reference outputs, so open-set leakage is a prominent limitation rather than a minor implementation detail
- the 100-household manuscript snapshot should be treated as a preview until larger frozen runs are published
- the discrepancy review is a qualitative developer audit, not an exhaustive external validation of every reference value
- benchmark success should not be interpreted as policy-advice readiness

The current paper is therefore an evaluation of model performance under a specific structured-output benchmark, not a general certification of tax or benefit competence.

## 9 Conclusion

PolicyBench shows a consistent pattern across both countries. Models often identify non-applicability, but positive tax and benefit amounts remain difficult, especially for multi-step income tax, payroll tax, National Insurance, and positive benefit cases. In the frozen manuscript snapshot, gpt-5.5 is the top-scoring model in both the US and UK.

These results support a narrow conclusion: unaided frontier models still struggle to reproduce selected household-level microsimulation outputs under a structured public benchmark. They do not show that models cannot assist policy analysis, and they do not validate PolicyEngine outputs as administrative truth. They suggest that future evaluations should separate no-tool estimation, tool-using system design, and reference-output validation more explicitly.

Next steps are to expand country coverage, increase frozen sample sizes, improve UK data provenance, and add protected or rotating evaluation sets so that public rankings are less



exposed to open-set leakage. The benchmark should also continue reporting sensitivity views, because equal-country global scores are useful summaries only when their weighting choices are visible.

## 10 Competing interests

The author is affiliated with PolicyEngine, which develops the microsimulation software used to produce the benchmark reference outputs.

Source: [Article Notebook](#)

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